

ZEPTO SQL Data Analysis Project

Project Summary

This project focuses on analyzing the product and inventory dataset of the popular e-commerce company ZEPTO using SQL Server. The main objective of this project is to perform data exploration, data cleaning, data manipulation, and business analysis using SQL queries.

The project helps in understanding product pricing, discounts, inventory availability, category-wise revenue, and product performance through structured SQL analysis.

Purpose of the Project

The purpose of this project is to:

- Explore and understand the ZEPTO product dataset
 - Clean and prepare the dataset for analysis
 - Perform business-oriented SQL analysis
 - Identify valuable business insights from product and inventory data
 - Improve SQL problem-solving and analytical skills
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Tools Used

- SQL Server Management Studio (SSMS)
 - Microsoft SQL Server
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Dataset Information

The dataset contains product-related information from ZEPTO including:

- Product Name
- Category
- MRP
- Discount Percentage
- Discounted Selling Price
- Available Quantity
- Product Weight
- Stock Availability

Step 1: Data Exploration

View Complete Dataset

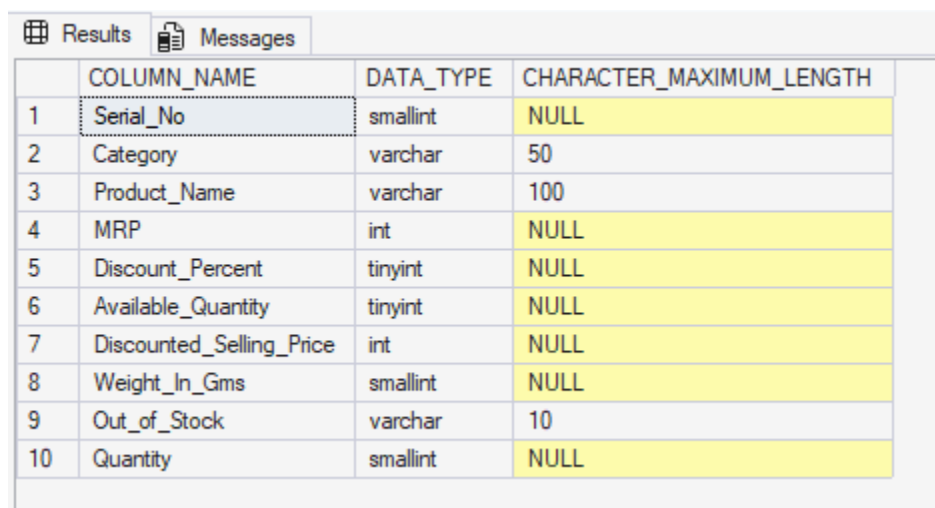
```
select * from zepto_data
```

Description

This query is used to display all records and columns from the dataset.

Check Column Information

```
select COLUMN_NAME,DATA_TYPE,CHARACTER_MAXIMUM_LENGTH  
from INFORMATION_SCHEMA.COLUMNS  
where TABLE_NAME like 'zepto_data'
```



The screenshot shows a database interface with a 'Results' tab selected. It displays a table with 4 columns: COLUMN_NAME, DATA_TYPE, and CHARACTER_MAXIMUM_LENGTH. The table contains 10 rows of data, with the first row highlighted in yellow. The columns are: Serial_No (smallint, NULL), Category (varchar, 50), Product_Name (varchar, 100), MRP (int, NULL), Discount_Percent (tinyint, NULL), Available_Quantity (tinyint, NULL), Discounted_Selling_Price (int, NULL), Weight_In_Gms (smallint, NULL), Out_of_Stock (varchar, 10), and Quantity (smallint, NULL).

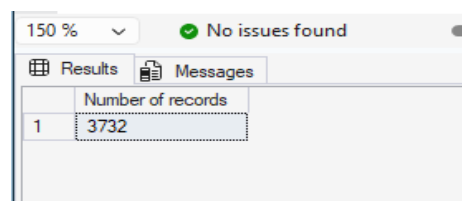
	COLUMN_NAME	DATA_TYPE	CHARACTER_MAXIMUM_LENGTH
1	Serial_No	smallint	NULL
2	Category	varchar	50
3	Product_Name	varchar	100
4	MRP	int	NULL
5	Discount_Percent	tinyint	NULL
6	Available_Quantity	tinyint	NULL
7	Discounted_Selling_Price	int	NULL
8	Weight_In_Gms	smallint	NULL
9	Out_of_Stock	varchar	10
10	Quantity	smallint	NULL

Description

This query is used to check the column names, data types, and character limits of the dataset.

Count Total Number of Records

```
select COUNT(*) [Number of records] from zepto_data
```



The screenshot shows a database interface with a 'Results' tab selected. It displays a table with 2 columns: Number of records and the count value. The table contains 1 row of data, with the first row highlighted in yellow. The columns are: Number of records (3732).

	Number of records
1	3732

Description

This query returns the total number of records present in the dataset.

Display Top 10 Records

select top 10 * from zepto_data

	Serial_No	Category	Product_Name	MRP	Discount_Percent	Available_Quantity	Discounted_Selling_Price	Weight_In_Gms	Out_of_Stock	Quantity
1	1	Fruits & Vegetables	Onion	25.00	16	3	21.00	1000	FALSE	1
2	2	Fruits & Vegetables	Tomato Hybrid	42.00	16	3	35.00	1000	FALSE	1
3	3	Fruits & Vegetables	Tender Coconut	51.00	15	3	43.00	58	FALSE	1
4	4	Fruits & Vegetables	Coriander Leaves	20.00	15	3	17.00	100	FALSE	100
5	5	Fruits & Vegetables	Ladies Finger	14.00	14	3	12.00	250	FALSE	250
6	6	Fruits & Vegetables	Potato	35.00	17	3	29.00	1000	FALSE	1
7	7	Fruits & Vegetables	Lemon	75.00	16	3	63.00	200	FALSE	200
8	8	Fruits & Vegetables	Watermelon	58.00	15	3	49.00	58	FALSE	1
9	9	Fruits & Vegetables	Capsicum Green	23.00	17	3	19.00	250	FALSE	250
10	10	Fruits & Vegetables	Chilli Green	19.00	15	3	16.00	100	FALSE	100

Description

This query displays the first 10 records from the dataset for quick inspection.

Step 2: Checking for Missing Values

Check Null Values in Dataset

```
select * from zepto_data  
where Category is null  
or  
Product_Name is null  
or  
MRP is null  
or  
Discount_Percent is null  
or  
Available_Quantity is null  
or  
Discounted_Selling_Price is null  
or  
Weight_In_Gms is null  
or  
Out_of_Stock is null  
or  
Quantity is null
```

Results

Messages

Serial_No	Category	Product_Name	MRP	Discount_Percent	Available_Quantity	Discounted_Selling_Price	Weight_In_Gms	Out_of_Stock	Quantity
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Description

This query checks whether any important columns contain NULL values.

Observation

The dataset does not contain any NULL values.

Step 3: Category Analysis

Display Unique Product Categories

```
select distinct Category from zepto_data
order by Category
```

Results		Messages	
	Category		
1	Beverages		
2	Biscuits		
3	Chocolates & Candies		
4	Cooking Essentials		
5	Dairy, Bread & Batter		
6	Fruits & Vegetables		
7	Health & Hygiene		
8	Home & Cleaning		
9	Ice Cream & Desserts		
10	Meats, Fish & Eggs		
11	Munchies		
12	Paan Corner		
13	Packaged Food		
14	Personal Care		

Description

This query displays all unique product categories available in the dataset.

Count Number of Categories

```
select COUNT(distinct Category) [Number of categories] from zepto_data
```

	Number of categories
1	14

Description

This query returns the total number of unique product categories.

Step 4: Stock Availability Analysis

Check Stock Availability

```
select Out_of_Stock,COUNT(Serial_No) [Stock Availability]
from zepto_data
group by Out_of_Stock
```

	Out_of_Stock	Stock Availability
1	FALSE	3279
2	TRUE	453

Description

This query checks how many products are available in stock and how many are out of stock.

Step 5: Product Purchase Analysis

Check Products Appearing Multiple Times

```
select distinct Product_Name,COUNT(Serial_No) [No of times purchased]
from zepto_data
group by Product_Name
having COUNT(Serial_No)>1
order by COUNT(Serial_No) desc
```

Results Messages		
	Product_Name	No of times purchased
1	Amul Delicious Fat Spread - Cholesterol Free	10
2	Arden Eggs White	10
3	Mother's Recipe Tamarind Paste	10
4	Quaker Oats	10
5	Saffola Veggie Twist Masala Oats	10
6	Sunfeast YiPPee! Magic Masala Noodles	10
7	Sunfeast Yippee! Pasta Treat - Sour Cream Onion	10
8	Kellogg's Real Almond & Honey Corn Flakes	9
9	Amul Fresh Cream	8
10	Sugarfree Gold Low Calorie Sweetner	8
11	iD Idli & Dosa Batter	7
12	Himalaya Anti Dandruff Shampoo	6
13	Godrej Yummiez Chicken Breakfast Salami	6

Description

This query identifies products that appear multiple times in the dataset. There are 1209 items purchased multiple times.

Step 6: Data Cleaning

Identify Invalid Product Pricing Records

```
select Product_Name,MRP,Discounted_Selling_Price
from zepto_data
where MRP = 0 and Discounted_Selling_Price = 0
```

Results Messages			
	Product_Name	MRP	Discounted_Selling_Price
1	Cherry Blossom Liquid Shoe Polish Neutral	0	0

Description

This query identifies records where both MRP and Discounted Selling Price are zero.

Delete Invalid Records

```
delete from zepto_data
where MRP = 0 and Discounted_Selling_Price = 0
```

Messages

(1 row affected)

Completion time: 2026-05-03T19:14:17.7641111+05:30

Product_Name	MRP	Discounted_Selling_Price
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Description

This query removes invalid pricing records from the dataset. And there is only one record is appropriate for the above query and that has deleted.

Verify Record Count After Cleaning

`select COUNT(*) [Number of records] from zepto_data`

Results	Messages
Number of records	
1	3731

Description

This query verifies the total number of records after deleting invalid entries. Here you one record has reduced.

Step 7: Data Transformation

Convert Paise into Rupees

`update zepto_data
set MRP = MRP / 100,
Discounted_Selling_Price = Discounted_Selling_Price / 100`

```
Messages

(3731 rows affected)

Completion time: 2026-05-03T19:20:08.5486491+05:30
```

Description

The original pricing values were stored in paise. This query converts the prices into rupees.

Display Updated Price Columns

select MRP,Discounted_Selling_Price **from** zepto_data

	MRP	Discounted_Selling_Price
1	25	21
2	42	35
3	51	43
4	20	17
5	14	12
6	35	29
7	75	63
8	58	49
9	23	19
10	19	16
11	29	24
12	11	9

Description

This query displays the updated price columns after conversion.

Change Data Type of Price Columns

alter table zepto_data **alter column** MRP **decimal**(8,2)

alter table zepto_data **alter column** Discounted_Selling_Price **decimal**(8,2)


```
Messages

(3731 rows affected)

Completion time: 2026-05-03T19:20:08.5486491+05:30
```

Description

This step changes the data type of price columns to decimal format for better readability.

Verify Updated Price Format

`select MRP,Discounted_Selling_Price from zepto_data`

	MRP	Discounted_Selling_Price
1	25.00	21.00
2	42.00	35.00
3	51.00	43.00
4	20.00	17.00
5	14.00	12.00
6	35.00	29.00
7	75.00	63.00
8	58.00	49.00
9	23.00	19.00

Description

This query confirms that the prices are displayed properly with decimal values.

Step 8: Business Questions and SQL Analysis

1. Find the Top 10 Best-Value Products Based on Discount Percentage

```
select distinct top 10 Product_Name,MRP,Discount_Percent
from zepto_data
order by Discount_Percent desc
```

Description

This query identifies the top 10 products offering the highest discount percentages.

Output Screenshot

	Product_Name	MRP	Discount_Percent
1	Dukes Waffy Chocolate Wafers	45.00	51
2	Dukes Waffy Orange Wafers	45.00	51
3	Dukes Waffy Strawberry Wafers	45.00	51
4	Ceres Foods Fish Mustard Instant Liquid Masala	220.00	50
5	Ceres Foods Laal Maas Instant Liquid Masala	220.00	50
6	Ceres Foods Nalli Nihari Instant Liquid Masala	220.00	50
7	Chef's Basket Durum Wheat Elbow Pasta	160.00	50
8	Chef's Basket Durum Wheat Fusilli Pasta	160.00	50
9	Chef's Basket Durum Wheat Penne Pasta	160.00	50
10	Dukes Waffy Chocolate Wafer Rolls	150.00	50

2. Find Products with High MRP That Are Currently Out of Stock

```
select Product_Name,MRP,Out_of_Stock
from zepto_data
where Out_of_Stock = 'true'
and MRP > (select AVG(MRP) from zepto_data)
order by MRP desc
```

Description

This query finds expensive products that are currently out of stock. There are exactly 73 items are out of stock.

Output Screenshot

70	ITC Master Chef Cheesy Corn Triangles	160.00	TRUE
71	Chef's Basket Durum Wheat Penne Pasta	160.00	TRUE
72	Safal Frozen - Sweet Corn	160.00	TRUE
73	ITC Master Chef Cheesy Corn Triangles	160.00	TRUE

✓ Query executed successfully.

3. Calculate Potential Revenue for Each Product Category

```
select Category,
SUM(Discounted_Selling_Price * Available_Quantity) [Total expected revenue]
from zepto_data
```

group by Category
order by [Total expected revenue] desc

Description

This query calculates the expected revenue generated from each product category.

Output Screenshot



	Category	Total expected revenue
1	Munchies	337369.00
2	Cooking Essentials	337369.00
3	Personal Care	270849.00
4	Paan Comer	270849.00
5	Packaged Food	224385.00
6	Chocolates & Candies	224385.00
7	Ice Cream & Desserts	224385.00
8	Home & Cleaning	122661.00
9	Health & Hygiene	64180.00
10	Dairy, Bread & Batter	55051.00
11	Beverages	55051.00
12	Biscuits	24995.00
13	Meats, Fish & Eggs	20693.00
14	Fruits & Vegetables	10846.00

4. Find Products Where MRP is Greater Than 500 and Discount is Less Than 10%

select distinct Product_Name,MRP,Discount_Percent
from zepto_data
where MRP > 500
and Discount_Percent < 10
order by MRP **desc**

Description

This query identifies premium products that have very low discounts. There are exactly 39 product items prices are more than 500 Rs. Maximum is 1250 and Minimum is 505

Output Screenshot

35	L'Oreal Paris Shine Masque Royal Jelly Hair Colour	550.00	0
36	Surf Excel Matic Front Load Detergent Powder	530.00	4
37	Amul Ghee (Tin)	520.00	2
38	Fresh Mutton Mince - Kheema (Halal)	515.00	3
39	Amul Pure Ghee (Pouch)	505.00	1

5. Identify the Top 5 Categories Offering the Highest Average Discount Percentage

```
select distinct top 5 Category,  
cast(AVG(Discount_Percent) as decimal(8,2)) [Avg discount percentage]  
from zepto_data  
group by Category  
order by [Avg discount percentage] desc
```

Description

This query identifies categories that provide the highest average discounts.

Output Screenshot

	Category	Avg discount percentage
1	Fruits & Vegetables	15.00
2	Meats, Fish & Eggs	11.00
3	Packaged Food	8.00
4	Biscuits	8.00
5	Ice Cream & Desserts	8.00

6. Find the Price Per Gram for Products Above 100g

```
select distinct Product_Name,  
Weight_In_Gms,  
Discounted_Selling_Price,  
CAST(Discounted_Selling_Price/Weight_In_Gms as decimal(10,2)) [price per gram]  
from zepto_data  
where Weight_In_Gms > 100  
order by Weight_In_Gms
```

Description

This query calculates the price per gram for products weighing more than 100 grams. There are exactly 1238 unique products are above 100g.

Output Screenshot

1234	Popular Essentials Rozana Sona Masuri Raw Rice	5000	312.00	0.06
1235	Saffola Gold (Jar)	5000	1240.00	0.25
1236	Super Saver Kolam Rice Sortex Clean	5000	630.00	0.13
1237	Aashirvaad Atta	10000	399.00	0.04
1238	Pillsbury Chakki Fresh Atta	10000	375.00	0.04

7. Group Products into Low, Medium, and Bulk Categories

```
select distinct Product_Name,  
Weight_In_Gms,  
case  
  when Weight_In_Gms <= 1000 then 'Low'  
  when Weight_In_Gms <= 3000 then 'Medium'  
  else 'Bulk'  
end [Weight in category]  
from zepto_data
```

Description

This query classifies products into Low, Medium, and Bulk categories based on weight. There are exactly 1781 unique products in the table.

Output Screenshot

1778	Zorabian Chicken Smoked Ham	250	Low
1779	Zorabian Pre Cut Chicken Skinless (Half)	500	Low
1780	Zucchini Green	250	Low
1781	Zucchini Yellow	250	Low

✓ Query executed successfully.

8. Calculate Total Inventory Weight Per Product Category

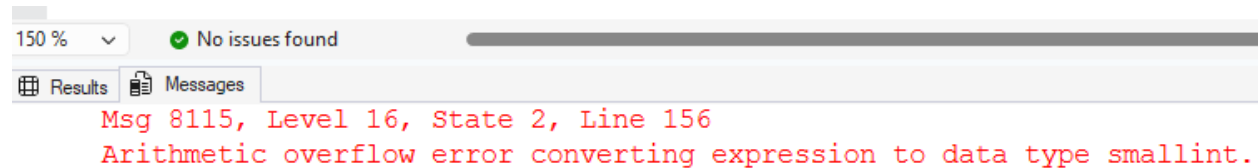
```
select distinct Category,  
SUM(Available_Quantity * Weight_In_Gms) [Total Inventory weight]  
from zepto_data  
group by Category  
order by [Total Inventory weight]
```

Description

This query calculates the total inventory weight for each product category.

Observation

Initially, this query produced an overflow error because the column data types were too small to handle large calculation outputs.



Change Data Type to Support Large Values

alter table zepto_data **alter column** Available_Quantity **int**

alter table zepto_data **alter column** Weight_In_Gms **int**

Description

This step changes the data type from SMALLINT to INT to support larger calculations.

Output Screenshot

	Category	Total Inventory weight
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Beverages	143735
6	Dairy, Bread & Batter	143735
7	Paan Comer	348187
8	Personal Care	348187
9	Home & Cleaning	373161
10	Packaged Food	490797
11	Ice Cream & Desserts	490797
12	Chocolates & Candies	490797
13	Cooking Essentials	1404654
14	Munchies	1404654

Project Conclusion

This project demonstrates practical SQL skills including:

- Data Exploration
- Data Cleaning
- Data Transformation
- Business Problem Solving
- Aggregate Functions
- Subqueries
- CASE Statements
- Data Type Handling
- Business Analysis using SQL

The project provides valuable business insights into product pricing, discounts, stock availability, and category-wise revenue analysis for the ZEPTO e-commerce dataset.

Future Improvements

- Create interactive dashboards using Power BI
- Automate reporting workflows
- Perform customer behavior analysis
- Add predictive analytics for demand forecasting